CLASSIFYING STACK OF A GROUP ALGEBRAIC SPACE IS ALGEBRAIC IFF THE GROUP IS FLAT LFP

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- General criterion specialization. lemma-group-quotient-algebraic (same chapter, named). Produces the covering B' and the flatness data.
- Descent curiosities. spaces-descent-lemma-curiosity (cross-chapter, named). Transfers flatness from the triple fibre product to the projection.
- Descent of properties. spaces-descent-lemma-descending-property-flat, spaces-descent-lemma-descending-property-locally-finite-presentation (cross-chapter, named). Descend flatness/lfp to G -> B.

Source reference. Original lemma 06PK: sources/stacks-criteria/pinned-original-source.tex, source lines 3385–3385. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NR: focused-seeds/geometry/S9-crit-BG-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 041W: focused-seeds/geometry/S9-crit-BG-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 041T: focused-seeds/geometry/S9-crit-BG-algebraic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 06FI: sources/stacks-criteria/pinned-original-source.tex, source lines 3245–3245. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme and let B be an algebraic space over S. Let G be a group algebraic space over B. Endow B with the trivial action of G. Then the quotient stack $[B/G]$ is an algebraic stack if and only if G is flat and locally of finite presentation over B.*

Proof. If G is flat and locally of finite presentation over B, then $[B/G]$ is an algebraic stack by Theorem 06FI.

Conversely, assume that $[B/G]$ is an algebraic stack. By Lemma 06PK and because the action is trivial, we see there exists an algebraic space B' and a morphism $B' \to B$ such that (1) $B' \to B$ is a surjection of sheaves and (2) the projections

$$B' \times_B G \times_B B' \to B'$$

are flat and locally of finite presentation. Note that the base change $B' \times_B G \times_B B' \to G \times_B B'$ of $B' \to B$ is a surjection of sheaves also. Thus it follows from Descent on Spaces, Lemma 06NR that the projection $G \times_B B' \to B'$ is flat and locally of finite presentation. By (1) we can find an fppf covering $\{B_i \to B\}$ such that $B_i \to B$ factors through $B' \to B$. Hence $G \times_B B_i \to B_i$ is flat and locally of finite presentation by base change. By Descent on Spaces, Lemmas 041W and 041T we conclude that $G \to B$ is flat and locally of finite presentation. □